

Sketching Behaviour Across a Career:

A Computational Analysis of 468 Le Corbusier Sketches

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1. About Data

This study addresses that gap by asking this question: How did Le Corbusier's sketching behaviour evolve from early to late career? The dataset compiled for this analysis (468 sketches drawn from all eight volumes of the Oeuvre Complete (1910-1969)) makes it possible to study this question at scale. By operationalising two proxy metrics derived entirely from the vectorized sketch geometry (move count and ink density), and situating them within William J. R. Curtis's three-period biographical framework, the analysis transforms a corpus of archival images into a longitudinal behavioural signal spanning 55 years.

2. Data Collection and Management Process

2.1 Source and Ethical Considerations

The dataset was assembled from Le Corbusier's Oeuvre Complete (8 vols., 1910-1969), accessed via the ITU EBSCO Library Database, so it is not an open-access data. Sketch selection was systematic within the available volumes rather than curated for a particular outcome. The three-period framework (Part I / II / III) follows Curtis (1987).

2.2 Dataset Structure

The dataset contains 468 rows, one per sketch, with eight columns. The two quantitative metrics ,move count and area covered (pixels), are the primary analytical variables. Content label (seven categories) and Era (year, used for decade binning) are the primary categorical variables. Period is derived from the folder structure, which follows Curtis's framework.

Column	Type	Description
filename	string	SVG filename (e.g. PART1 (10).svg)
folder	string	Source folder path by part
period	string	Full period name string (Curtis framework)
content_label	string	Manually assigned content category (7 types)
move	integer	SVG path count — proxy for discrete pen gestures
Area Covered (Pixels)	integer	Non-white pixel count — proxy for ink density
Era	integer	Year extracted from filename context

2.3 Cleaning and Processing

The dataset was assembled through a five-step pipeline: (1) manual screenshot extraction of sketch pages from the Oeuvre Complete PDFs; (2) crop and resize to 256x256 px PNG with

background cleaning; (3) manual content labeling via an interactive Google Colab tool; (4) PNG-to-SVG vectorization (Potrace, outline mode, threshold=200), with move count = count of SVG path elements, (5) ink density computed as non-white pixel count divided by 65,536 (total pixels in a 256x256 canvas).

The dataset arrived complete: zero missing values across all eight columns, zero duplicate rows, zero filename conflicts. The only analytical transformation was the consolidation of art, object, and other labels into a single other category for period-level comparisons, preserving the full nine-category breakdown for content-specific analysis.

3. Analysis Method

Two quantitative metrics were computed for each sketch. Move count is defined as the total number of SVG path elements after vectorization, each path corresponds to a discrete pen-lift in the original drawing. Ink density (occupancy) is defined as the proportion of non-white pixels in the source PNG, representing the spatial coverage of each sketch.

Part-level comparisons use mean and median values for both metrics, with standard deviation reported to capture within-period variability. Content label distributions are reported as percentages within each period to control for the unequal sample sizes (Part I: 72; Part II: 184; Part III: 212). The increasing sample size across parts reflects the Oeuvre Complete's editorial structure and is treated as a feature of the source material rather than a sampling bias.

Era-level analysis bins sketches into six groups (1910s through 1960s) and computes mean move count per decade to identify finer-grained temporal patterns within the three-period framework. Outlier detection uses IQR-based boxplots and Z-score thresholds ($|z| > 3$), identifying 28 move-count outliers and 21 area outliers; all are retained as legitimate intensive sketching sessions rather than measurement errors.

4. Visualisation and Findings

The output of this study is a two-tab interactive HTML visualisation: Dataset Info and Sketch Visualization. All colors are drawn from Le Corbusier's own Salubra 1931 wallpaper palette.

The Dataset Info tab presents: the central research question and part statistics; three period cards with key metrics; the full data quality report across seven sub-tabs; the color keyboard showing the Salubra palette mapping; and the dataset construction pipeline diagram. The Sketch Visualization tab displays all 468 sketches as a navigable grid with interactive hover metadata, five-channel stripe encoding (part, move count, occupancy, content label, era), and toggle-able move-line overlays.

Key Findings

Metric	Part I (1887-1922)	Part II (1922-1944)	Part III (1945-1965)
Sketches (n)	72	184	212
Mean move count	99.4	78.3	113.8

Median move count	79	50	76
Std move count	~90	~75	~120
Mean ink density	20.9%	68.8%	89.2%
Career change (I→III)	—	—	Move: +15% · Ink: x4.3

Ink density is the dominant career signal. Occupancy grew from 20.9% to 89.2% ,a 4.3x increase, while move count changed only +15%. Le Corbusier made approximately the same number of gestures throughout his career, but each gesture covered progressively more of the page.

Part II compression is more extreme than anticipated. Move count dropped to a mean of 78.3 (median 50) while ink density tripled relative to Part I. The sharpest median-to-mean gap of any period (50 vs. 78.3) signals high internal variance: most Purist sketches are economical, but a subset is extremely dense. This coincides with the Purist period which is economy of line, density of intent.

Part III recovery reaches career-high complexity. Move count recovered to its career high (mean 113.8, std ~120) while ink density continued rising to 89.2%. The decade-level data reveals that the 1950s alone show the highest mean move count of any decade (126.8), corresponding to the Chandigarh construction period. The increasing standard deviation in Part III suggests that late-career sketching became simultaneously more expansive and more variable.

Scene almost disappears in Part II, then partially recovers. Scene sketches collapse from 16.7% of Part I to just 3.3% of Part II. Section rises from 2.8% to 13.6% in Part II and remains elevated in Part III (13.7%), reflecting the shift from naturalistic observation to architectural section thinking.

Art emerges only in Parts II and III. Art as a content category is entirely absent from Part I (0.0%), rises to 2.2% in Part II, and reaches 5.7% in Part III. This suggests that the late LC was doing something with the sketch that exceeded pure architectural instrumentality.

Content Label Distribution by Period

Label	Part I	Part II	Part III	Total
landscape	30.6% (22)	27.2% (50)	24.5% (52)	124
plan	20.8% (15)	25.0% (46)	17.5% (37)	98
building	16.7% (12)	16.8% (31)	19.3% (41)	84
section	2.8% (2)	13.6% (25)	13.7% (29)	56
scene	16.7% (12)	3.3% (6)	8.5% (18)	36
other*	6.9% (5)	9.8% (18)	6.1% (13)	36
art	0.0% (0)	2.2% (4)	5.7% (12)	16

facade	4.2% (3)	1.1% (2)	2.4% (5)	10
object	1.4% (1)	1.1% (2)	2.4% (5)	8

* other consolidates art, object, and other labels for period-level analysis.

5. Conclusion

This study demonstrates that computational sketch analysis can detect systematic behavioural shifts across a 55-year architectural career. The central finding (that move count remained nearly stable (+15%) while ink density quadrupled (x4.3)) provides a quantitative basis: that Le Corbusier's later career was characterized by a deepening of presence rather than a simplification of means. The same hand, making roughly the same number of gestures, committed progressively more of itself to each page.

The content label analysis adds a second dimension to this narrative. The collapse of scene drawing in Part II and the simultaneous rise of section suggests that the Purist period involved not just a change in how Le Corbusier drew, but a change in what modes of seeing the sketch was asked to perform. The partial recovery of scene in Part III, alongside the emergence of art as a category, suggests that the late career involved a partial return to the observational and exploratory modes that the middle period had suppressed.

The visualisation translates these findings into an interactive platform that allows researchers and students to navigate the full corpus, inspect individual sketches, and compare periods through multiple encoding channels.

Limitations

- Move count is a proxy, not a direct measure. SVG path count reflects Potrace's vectorization decisions as well as Le Corbusier's pen gestures.
- Sample size increases monotonically across parts (72, 184, 212). This reflects the editorial structure of the Oeuvre Complete.
- Content labeling is manual and categorical. The seven-category scheme imposes discrete boundaries on a continuous spectrum of sketch types.

Despite these limitations, the core findings are robust: the ink density trajectory and the scene-to-section content shift are visible across multiple analytical approaches. The visualisation preserves the full distribution, including outliers, allowing readers to assess the strength of the patterns directly.

References

Curtis, W. J. R. (1987). *Le Corbusier: Ideas and Forms*. Phaidon Press.

Le Corbusier. (1910-1969). *Oeuvre Complete* (8 vols.). Artemis Verlag. [Accessed via ITU EBSCO Library Database.